MANVIK TALWAR

AI Engineer | Machine Learning, LLM Applications & Model Deployment
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SUMMARY

Final-year B.Tech CSE (AI & ML) student with two AI engineering internships shipping ML and LLM systems for enterprise use. Works end to end: data preprocessing, model training and evaluation, REST API integration, cloud and on-prem deployment, monitoring and automated testing, with a focus on reliable, responsible AI.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript
Machine Learning: NumPy, pandas, scikit-learn, PyTorch, TensorFlow/Keras, CNNs, OpenCV, MediaPipe, Anomaly Detection
Generative AI: LLMs, RAG, LangGraph, LangChain, Agentic AI, MCP, Prompt Engineering, Hugging Face, llama.cpp, Ollama, Embeddings
Backend & Data: FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, MySQL, Redis, ChromaDB, pgvector, SQLAlchemy, Pydantic
Cloud & MLOps: Azure OpenAI, Azure Cognitive Search, Docker, Kubernetes, Prometheus, Grafana, GitHub Actions CI/CD, pytest, Git
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Statistics & Probability, Linear Algebra

PROFESSIONAL EXPERIENCE

AI Engineering Intern | Niva Bupa Health Insurance **May 2026 – Jul 2026**

- Built a **claims auto-adjudication engine**: a 7-gate pipeline (policy, member, coverage, waiting period, exclusions, financials, state) pairing deterministic business rules with an **LLM for clinical judgments**, run **on-prem** (Gemma, llama.cpp) or on cloud.
- Made LLM output production-safe with **grammar-constrained decoding** (GBNF) and Pydantic schema validation; **4-tier confidence routing** sends uncertain claims to operations, clinical or senior-auditor queues instead of an automated verdict.
- Cut LLM calls per bill line item from **~11 to 1** via batched screening and caching; served the engine as an async **FastAPI** API with per-gate SSE streaming to a React dashboard and OCR + LLM document extraction; **77 automated tests**, incl. property-based.

AI Development Intern | Path Infotech **Jun 2025 – Aug 2025**

- Built and deployed an **enterprise RAG chatbot** on company servers answering staff questions from internal policy and HR PDFs, using **Azure OpenAI**, **Azure Cognitive Search** (custom PDF indexing skillsets) and Blob Storage, with live indexing of new uploads.
- Added a **citation layer** (Flask API + MySQL tagging) so every answer links to its source document, keeping responses auditable.

PROJECTS

IntelliOps: Agentic AIOps Platform | Deloitte Capstone 2026 | *FastAPI, Redis, Kubernetes* **github.com/CodexManvik/intelliops**

- Built a **7-service event-driven platform** (FastAPI, Redis Streams, PostgreSQL) that groups alert storms into incidents, ranks root causes and runs **reversible remediation on Kubernetes** with automatic rollback; logs, metrics and audit trail in Prometheus, Grafana and Loki.
- Implemented pluggable **anomaly detectors** (online z-score, robust median/MAD, Isolation Forest) with a seeded benchmark of precision, recall and false-positive rate; the robust detector cut seasonal false positives **7x**, and CI blocks regressions.
- Limited the LLM agent to a **closed set of remediation actions** behind a fail-closed human-approval gate; playbooks graduate to automatic only after a record of verified fixes, and outcomes feed back as training data. **860+ tests**.

Interview Mirror: AI Interview Coach | Team of 2 | *React, FastAPI, Supabase* **interview-mirror-hy2n.vercel.app**

- Deployed an AI interview simulator with a 3-person panel (technical, manager, HR) driven by the candidate's resume and target JD; Whisper speech-to-text, and body language read **on-device** (MediaPipe) so video never leaves the browser.
- Designed **evidence-based scoring** (two LLM graders, tie-break, transcript evidence check, code-computed scores) and an **LLM reliability layer** with model fallbacks, timeouts and rate-limit cooldowns; **950+ backend and 300+ frontend tests** in CI.

Aethel: Local-First AI Agent for Windows | *FastAPI, Tauri, React, llama.cpp, MCP* **github.com/CodexManvik/Aethel**

- Shipped a desktop agent (v2.0 installer) that runs multi-step PC tasks through an **MCP tool hub** with permission manifests, approval gates and rollback; learned skills compile into **LLM-free macros** after 3 identical runs.
- Integrated a **CPU-only decision model (~0.2 s/call)** for intent routing: **0.91 precision/recall** on held-out labelled data. **790+ tests**.

FloatChat-AI: NL Ocean Data Explorer | SIH 2025 | *pandas, PostgreSQL, ChromaDB, Docker* **github.com/CodexManvik/FloatChat-AI**

- Built a pipeline that interpolates Argo float profiles onto 24 standard depth levels and loads **88K+ measurements** into PostgreSQL and a vector index; questions route to **NL-to-SQL** (validated read-only) or filtered semantic search, answered with maps and charts.

Deep Learning Image Watermarking | *TensorFlow, Keras* **github.com/CodexManvik/Deep-Learning-Based-Watermarking**

- Trained a differentiable encode-attack-decode **CNN (4.1M params, ~18 ms/image)** hiding a 256-bit payload in the DWT domain; on 50K images: **37.3 dB PSNR, 0.97 SSIM, 3.2% bit error**, 2.4–7x lower than a classical baseline; paper in preparation.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) | Manipal University Jaipur | CGPA: 7.91/10 **Aug 2023 – Aug 2027**

Class XII: 81.8% (2023) | Class X: 89.8% (2021)

ACHIEVEMENTS

- **Smart India Hackathon 2025 National Semi-Finalist** (FloatChat-AI) | Dell x GDG Ideathon Finalist | 4x Student Excellence Award, MUJ | NPTEL Topper: Deep Learning; Data Structures & Algorithms